

Report on the UNICEF data set

Introduction

UNICEF represents a fund of the United Nations meant to distribute developmental aid to impoverished children worldwide. It is usually targeted to children from regions that are underdeveloped economically, experience extreme weather conditions, and are affected by war.

This data set contains observations of 121 countries on various variables spanning from child mortality rate to foreign aid as a percentage of gross domestic product. The data was pre-processed and instead of the original 146 countries, there are 121 countries. The set of variables present in our data set is a subset of the larger set of variables. The data was collected and organized into this data set by UNICEF in the year 1997 and is part of the larger data set called “The State of the World’s Children 1997”.

This report could be useful to scientists to provide further explanation of child mortality. It should relate child mortality to the set of variables and give social scientists the possibility to further understand how this rate could be lowered. It could also be used as a tool that provides insight to policymakers into the type of strategy that can be most effective when combating the child mortality rate. The second goal is to build a model suitable for predicting future values of child mortality rate. The report should also contain any anomalies considering the data.

Data visualization

To begin with, we have a correlation matrix representing the correlation coefficients between variables. The correlation in turn represents the relation between variables such that, the further the coefficients are from 0 the stronger the linear relation between the two. The warmer colors represent the values above, and the colder ones below zero. There are a couple of interesting points we should name.

First of all, there is a negative correlation between each of the variables and mortality aside from foreign aid. This means that the correctly directed aid could potentially cause the lowering of mortality rate. This is however not certain, as the correlation does not explain the causal relationship between the two variables.

The second important thing is the very high correlation between the two literacy variables. Which if not controlled properly could cause collinearity problems. This will portray that the two variables are significant when estimating the model when they just explain each other instead.

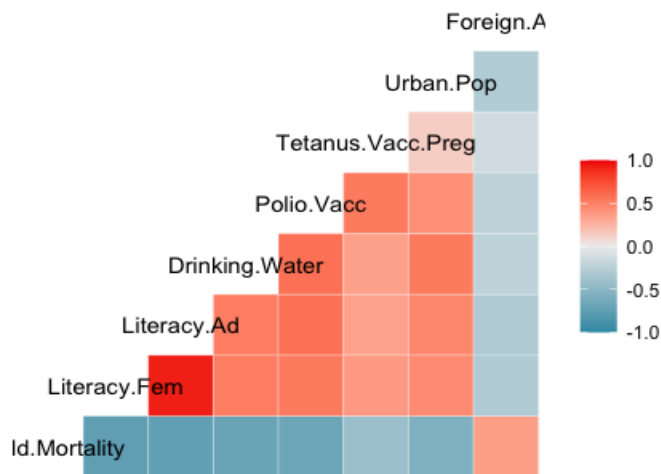


Figure 1: Correlation Matrix

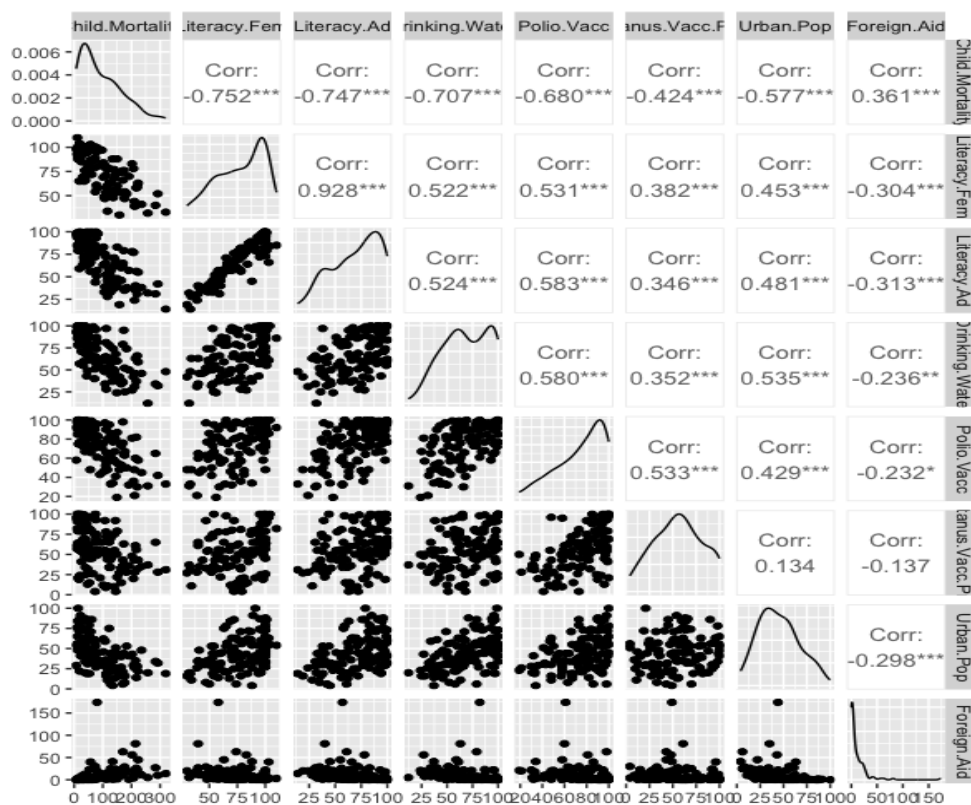


Figure 2: Pairwise Comparison of Multivariate Data

The second graph could be split into three parts. The diagonal entries are the distribution of each of the variables. This is not of interest to us as there are no previous assumptions regarding their distributions, and their differences should not affect the model. The upper triangular entries are numerical writing of the previous graph and could be easier to read when interpreting the relationship. The lower triangular entries are graphical

linear relations between two variables. We could notice when considering foreign aid, that there exists a certain potential outlier which shall be explored in further analysis. We could also again speculate about a very close relationship between two literacy variables, which may lead us to include only one in the model, so we could bypass collinearity.

Statistical Analysis

Regarding the statistical analysis part of the report, I tried to apply different statistical methods, while looking for a possible model, which shall bear the answer to the previously stated question. Explaining the effect of variables on the child mortality rate.

As expected, I first tried fitting the regular least squares model, with all the observations including possible outliers and all the variables. The conclusion was that the variable with the most significant effect on child mortality was drinking water, closely followed by the polio vaccine, and later with female literacy and urban population. According to adjusted R^2 the model was able to explain roughly 74% of variability, which for a basic model seems quite promising. The next logical step is observing the model diagnostics and seeing if the model is satisfying the assumption. Which will make our results valid.

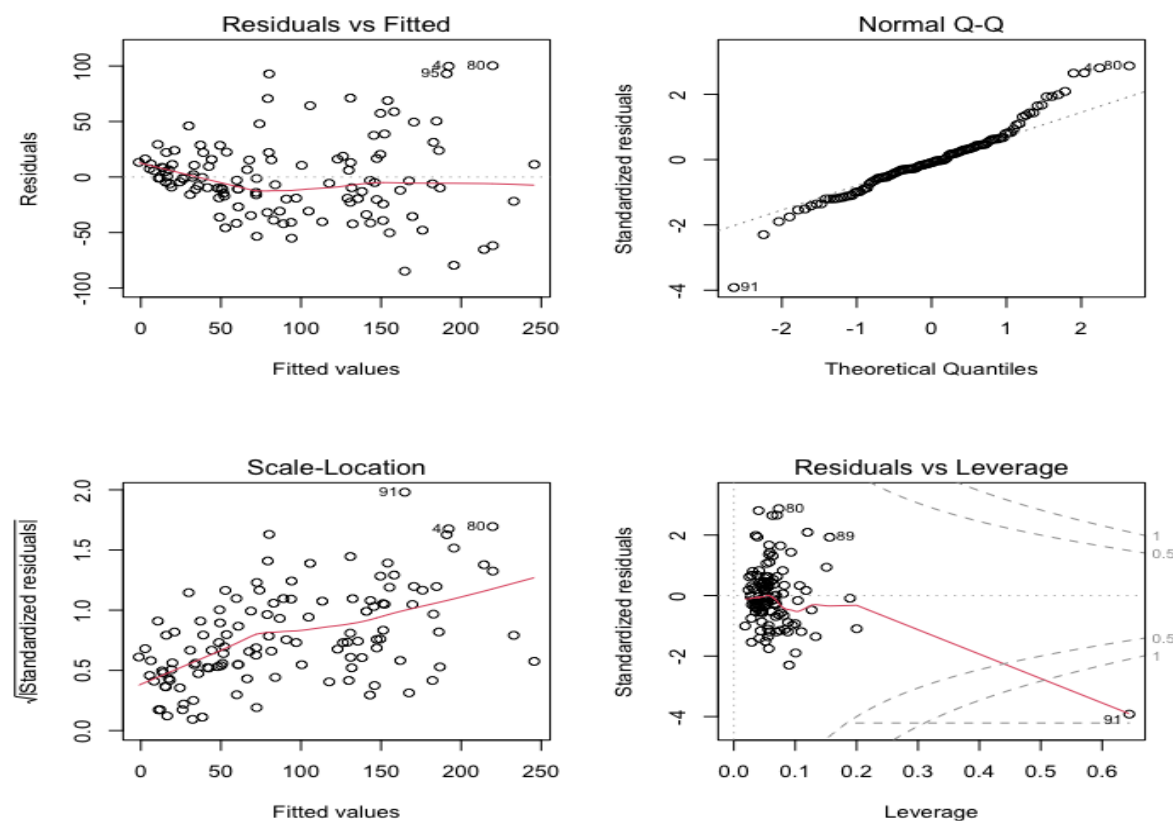


Figure 3: Model Diagnostics

There is quite a lot of statistical theory behind these plots which we will not delve into. For us, the important thing is that these four plots will allow us to verify the assumptions behind our model. According to the Residuals vs Fitted plot, our model linear assumptions hold, as the running mean (red line) stays around zero. If we had taken a more conservative stance both the Normal Q-Q plot and Scale-Location plot would have rejected our assumption. However, for the sake of the report and considering that this is the usual case in real data we are going to consider that the homoscedasticity and normality assumption hold. The thing of interest to us is the fourth plot, in which we can see that there is an outlier in our data. The observation 91, or São Tomé and Príncipe. Further research will tell us if this influential outlier is a good or bad leverage point.

In Figure 4, we can see plots of fitted regression lines, between any available variable and child mortality. In each of the scatter plots, we could spot some outliers although masked. However one, regarding foreign aid to São Tomé and Príncipe stands out. This might be due to a few factors. At the time of data collection, Sao Tome was a relatively new country, and UNICEF with additional foreign aid tried to stir the country into ratifying laws protecting children. Sao Tome also has the second smallest economy in Africa, making poverty the main issue when it comes to the well-being of children, to which the UN responded with foreign aid through UNICEF. Since the GDP of a country is so small, a percentage of GDP as foreign aid might be exaggerated.

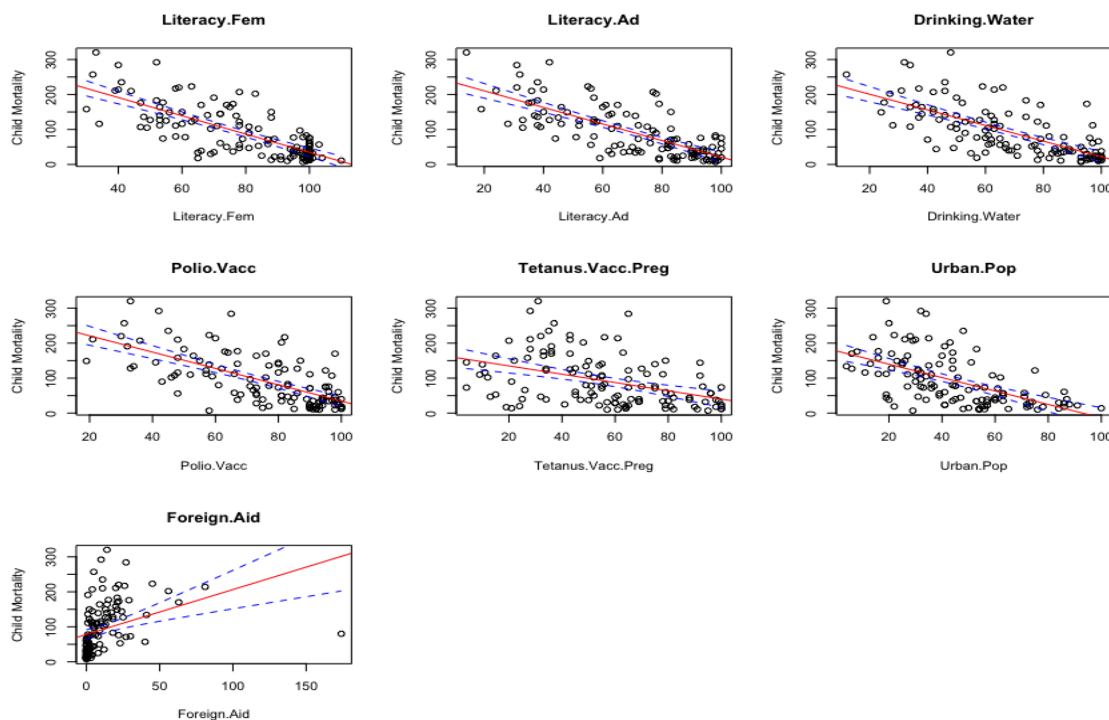


Figure 4: Individual Linear Regression

The next step shall be cross-validation. We try to estimate errors and compare models based on training and test data. The loo-cv, over 7 non-pre-processed variables 121 times will be the procedure I used. After applying this procedure, we see that square loss, is higher than residual standard error. Meaning that the outlier has too much of a negative effect on the model. The outlier was chosen based on the value of the mean square error. Observation 91 had the biggest MSE, and I subsequently removed it. It was São Tomé and Príncipe. A newly fitted model proved better, as it explained 78% of variability, and foreign aid proved a significant regressor, the significance of other regressors increased too.

However, not removing the outlier and instead assigning it less weight, might give us more precise estimators, and let us keep the integrity of our data, as we do not reduce sample size. This will be achieved through robust regression as we apply MM estimator. It proved as a better method, not only because it explained more variability, but also because it controlled for other outliers that we previously did not account for.

After this, I also assessed the previous problem of collinearity. Namely, I removed adult literacy. This did not prove as a problem, because the explained variability got smaller, and the significance was not affected. Hence we left it inside our data set for future models.

As I final thing, I compared predictions obtained from a regular OLS model, robust model, and actual values in figure 5. According to the mean square error, a robust model comes out as best when it comes to predicting child mortality

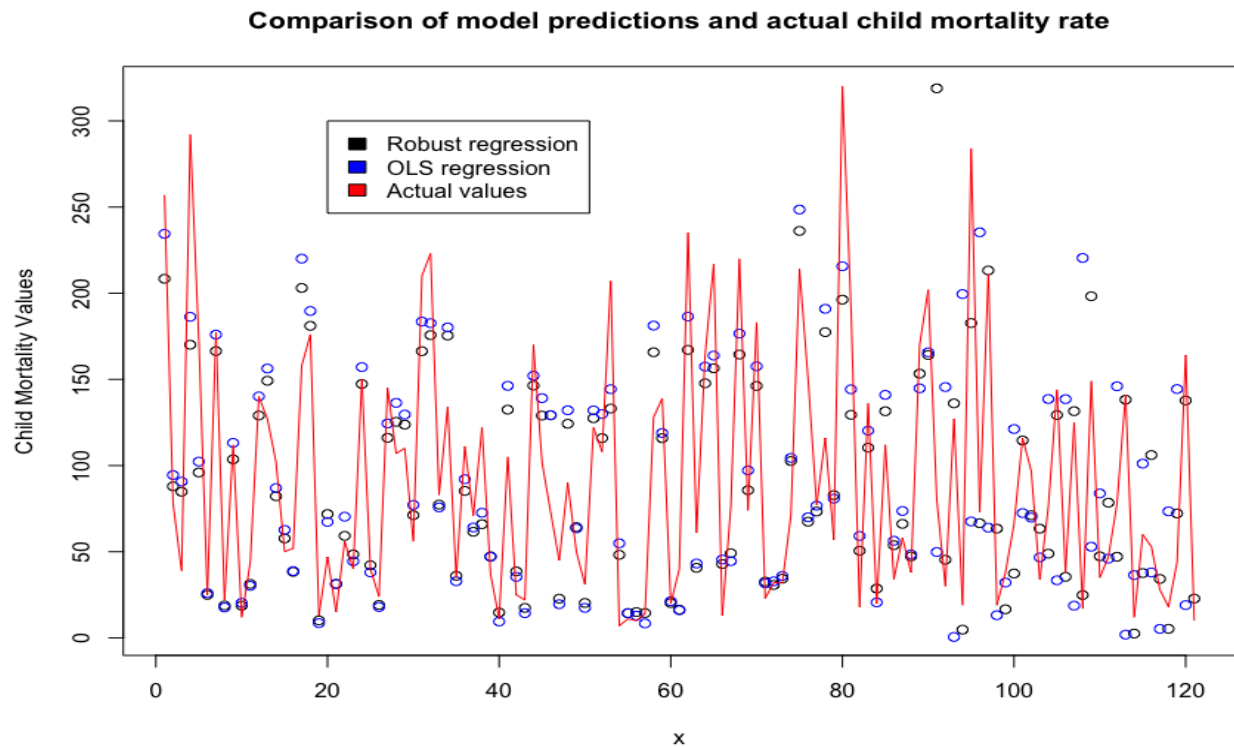


Figure 5: Comparison of Models

Particular Features of the Data

Regarding certain features of the data, we should once again mention the outlier which is Sao Tome and Principe, which was treated using robust regression. The byproduct of which, was also treating masked outliers of Angola and Niger, and assigning appropriate weights to each of them.

Conclusion for the Client

The final finding of the report suggests that the best model for predicting child mortality is a robust regression model. The variables that have the biggest impact on child mortality are foreign aid, followed by drinking water and literacy of females.

Appendix

```
unicef <- read.table("unicef_nonmissing.dat",header=TRUE,stringsAsFactors=FALSE)

install.packages("GGally")

library("GGally")

#####

# Graphical Analysis

#####

# 1st graph

ggcorr(unicef)

# 2nd graph

unicef1 <- unicef[,-1]

ggpairs(unicef1)

#####

# Statistical Analysis

#####

# OLS model

regressors <- colnames(unicef)

regressors <- regressors[-c(1,2)]

mortality <- lm(Child.Mortality ~ Literacy.Fem + Literacy.Ad + Drinking.Water +

               Polio.Vacc + Tetanus.Vacc.Preg + Urban.Pop + Foreign.Aid,

               data = unicef)

summary(mortality)

par(mfrow = c(2,2))

plot(mortality)

par(mfrow = c(3,3))

for (i in regressors) {
```

```
plot(unicef[[i]],unicef$Child.Mortality, main = i, xlab = i, ylab = "Child Mortality")

model <- lm(as.formula(paste("Child.Mortality ~",i)), data = unicef )

newx <- seq(min(unicef[[i]]),max(unicef[[i]]))

preds <- predict(model, newdata= setNames(data.frame(newx), i), interval = "confidence")

abline(model, col = "red")

lines(newx, preds[,3], lty = "dashed", col = "blue")

lines(newx, preds[,2], lty = "dashed", col = "blue")

}

# Running the LOO-CV

n <- 121 # Number of observations

yhat <- numeric(0)

for (i in 1:n){

  fmi <- lm(Child.Mortality ~ Literacy.Fem + Literacy.Ad + Drinking.Water +

    Polio.Vacc + Tetanus.Vacc.Preg + Urban.Pop + Foreign.Aid, data=unicef[-i,]) # Fit without point i

  yhat[i] <- predict(fmi,unicef[i,]) # Prediction of y_i

}

sqloss <- sqrt(mean((yhat-unicef$Child.Mortality)^2))

l1loss <- mean(abs(yhat-unicef$Child.Mortality))

#max(yhat)

# yhat[[91]]

# We exclude the 91st observation which is São Tomé and Príncipe

unicef <- unicef[-91,]

# Now we do regression and check for coefficients

model1 <- lm(Child.Mortality ~Literacy.Fem + Literacy.Ad + Drinking.Water +

  Polio.Vacc + Tetanus.Vacc.Preg + Urban.Pop + Foreign.Aid,

  data = unicef )

summary(model1)

# We do robust regression and apply MM estimator
```

```
installed.packages("robustbase")

library(robustbase) # This is needed for lmrob!

lm3mm <- lmrob(Child.Mortality~Literacy.Fem + Literacy.Ad+Drinking.Water +
               Polio.Vacc + Tetanus.Vacc.Preg + Urban.Pop + Foreign.Aid,data=unicef) # MM-estimator

par(mfrow=c(2,3))

plot(lm3mm)

summary(lm3mm)

install.packages("car") # If not installed

library(car)

unicef <- read.table("unicef_nonmissing.dat",header=TRUE,stringsAsFactors=FALSE)

lm_temp <- lm(Child.Mortality~Literacy.Fem + Literacy.Ad + Drinking.Water +
              Polio.Vacc + Tetanus.Vacc.Preg + Urban.Pop + Foreign.Aid, data = unicef) # Standard regression

vif(lm_temp) # Check variance inflation factors

unicef <- unicef[,-4]

lm3mm2 <- lmrob(Child.Mortality~Literacy.Fem + Drinking.Water +
               Polio.Vacc + Tetanus.Vacc.Preg + Urban.Pop + Foreign.Aid,data=unicef) # MM-estimator

summary(lm3mm2)

###

#Prediction

par(mfrow =c(1,1))

unicef <- read.table("unicef_nonmissing.dat",header=TRUE,stringsAsFactors=FALSE)

y_hat1 <- predict(lm3mm, newdata = unicef)

x <- seq(1,121,1)

plot(x, y_hat1, ylab = "Child Mortality Values", main = "Comparison of model predictions and actual child
mortality rate")

lines(x,unicef$Child.Mortality, col = "red")

unicef <- unicef[,-91,]
```

```
y_hat2 <- predict(model1,newdata = unicef)
```

```
x1 <- seq(1,120,1)
```

```
points(x1,y_hat2, col = "blue")
```

```
legend(20, 300, legend=c("Robust regression", "OLS regression", "Actual values"),
```

```
      fill = c("black","blue","red")
```

```
)
```

```
MSE1 <- mean((y_hat1-unicef$Child.Mortality)^2)
```

```
MSE2 <- mean((y_hat2-unicef$Child.Mortality)^2)
```